

BASE GUIDE 1 OF 3

WHAT We Are Doing

Exact research question, scope, comparison, and result space

Cross-Cycle Explanation Drift in Solar Flare Forecasting

Version 2.0 | October 2026 | GitHub Base Research Documentation

Status	REVIEWED / BASELINE LOCKED
Research domain	Space Weather Machine Learning Explainable AI

1. Research in One Sentence

PLAIN-LANGUAGE SUMMARY

Hum Cycle 24 ke definitive SHARP data par flare-prediction models train karenge, unhe freeze karenge, aur same pipeline ko Cycle 24 holdout aur Cycle 25 temporal blocks par chalayenge. Hum prediction performance ke saath SHAP attribution patterns compare karenge, phir controls se test karenge ki observed explanation difference normal variability hai ya meaningful cross-cycle drift.

2. What Exactly Is Being Tested?

The research has two parallel outputs for the SAME frozen model:

Axis	Question	Primary evidence
Predictive stability	Does forecasting skill change from Cycle 24 holdout to Cycle 25?	TSS + confidence intervals, complementary metrics, calibration
Explanation stability	Does the distribution/structure of SHAP attributions change across the same evaluations?	Mean SHAP , sign consistency, rank correlations, top-k overlap, distribution-distance statistics, null controls

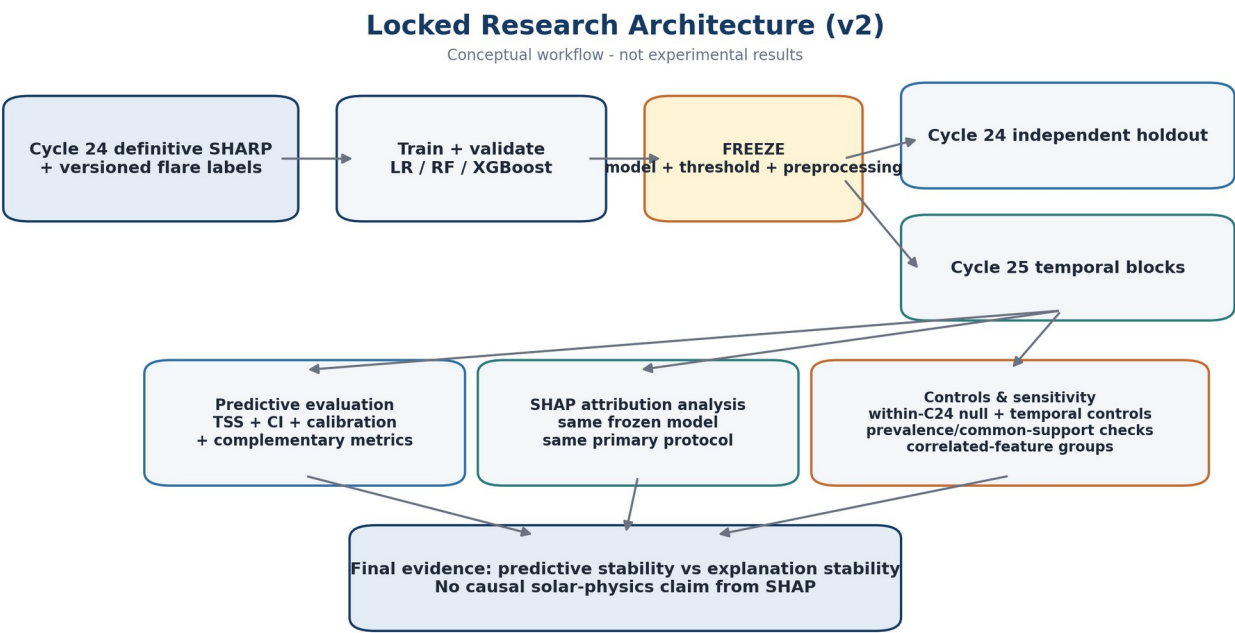


Figure 1. What the project compares. This is a conceptual design, not a result.

3. What Stays Fixed vs What Changes

Fixed by design	Allowed to change / be measured
Feature definition and primary SHARP product	Input feature distributions across cycles
Cycle-24-fitted preprocessing	Flare prevalence across blocks
Trained model parameters	Predictive performance
Classification threshold rule	SHAP attribution distribution
Primary SHAP explainer settings/background	Calibration and error profile
Target definition and event-association rule	Active-region population characteristics

Important precision

The frozen model does not "change internally" on Cycle 25. Its parameters remain fixed. What can change is the prediction behavior and the distribution of SHAP attributions when the model receives a different temporal/domain input distribution.

4. Data Scope

Component	Current locked direction
Magnetic data	Definitive SDO/HMI SHARP; exact JSOC series locked and versioned before extraction (preferred research series: hmi.sharp_cea_720s).
Native cadence	12 minutes; research sampling stride must be pre-registered before final model fitting.
Training window	May 2010 - December 2019 (Cycle-24-era HMI/SHARP window).
Evaluation window	2020 onward in pre-defined Cycle-25 temporal blocks.
Target	$\geq$ M-class flare in the next 24 hours, with a versioned event-association rule.
Models	Logistic Regression, Random Forest, XGBoost.

5. What Counts as Explanation Drift?

A raw SHAP ranking difference is not enough. We call the result evidence for cross-cycle explanation drift only if a pre-specified drift statistic is unusually large relative to within-Cycle-24 controls and remains robust to the planned sensitivity analyses.

Evidence Ladder Before Declaring Explanation Drift

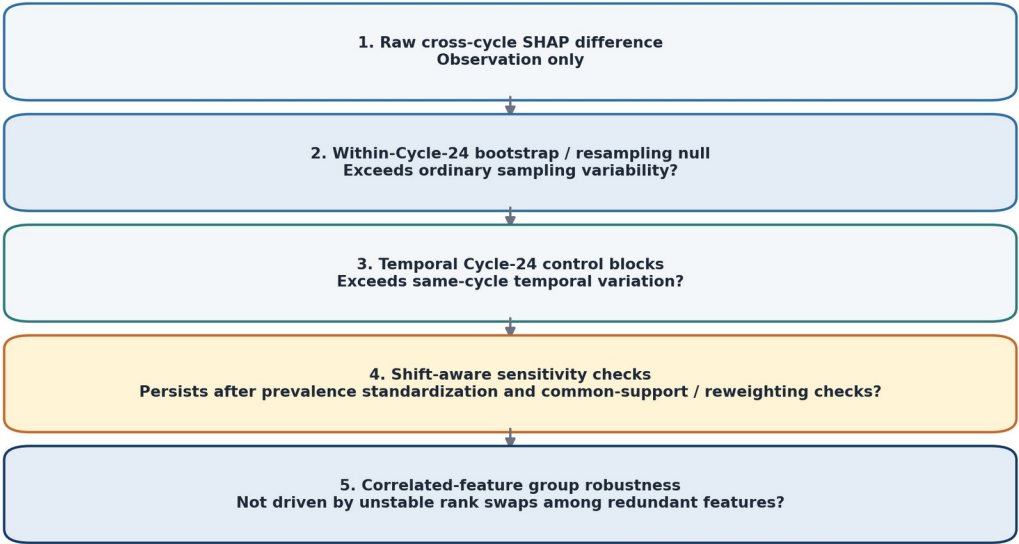


Figure 2. Evidence required before using the term explanation drift.

6. Possible Results

Pre-Registered Interpretation Matrix

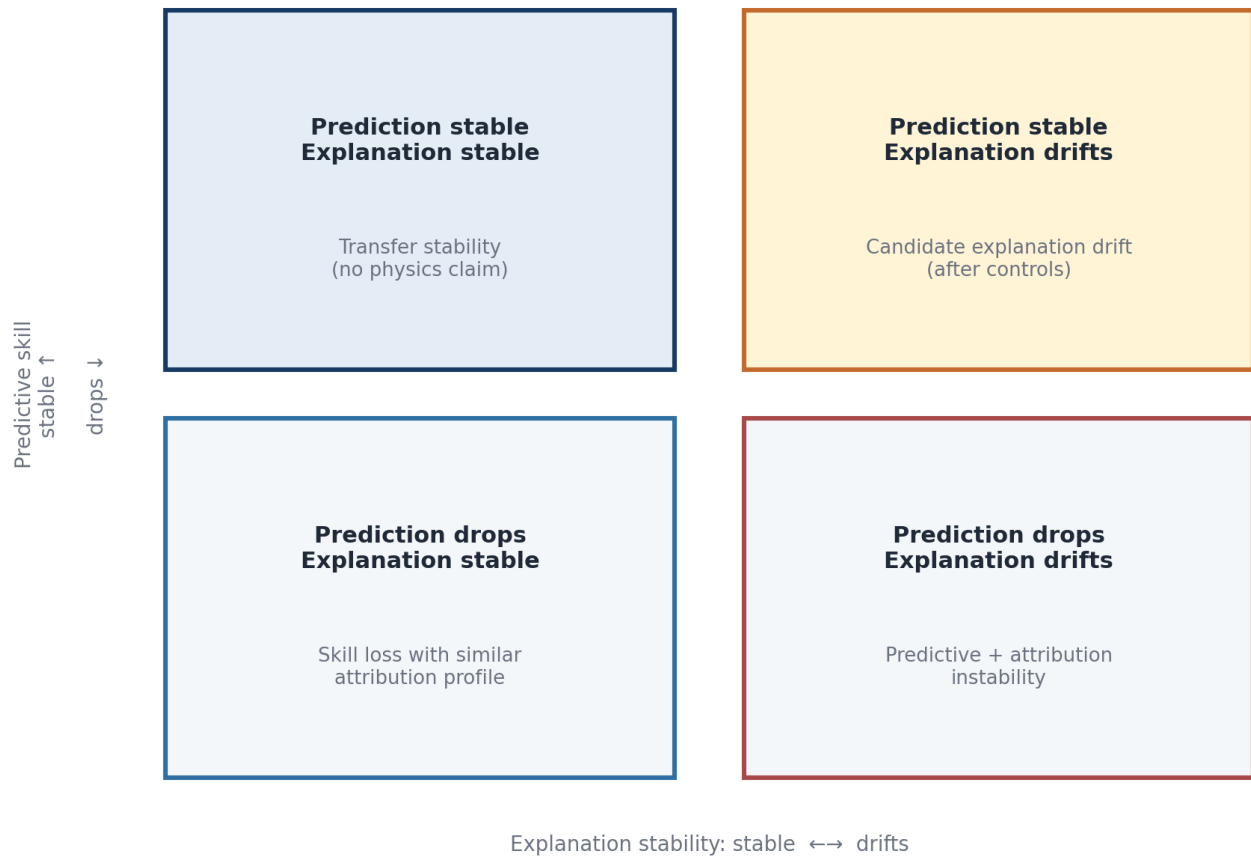


Figure 3. All four outcome families are scientifically valid.

7. What This Research Is NOT

- Not a claim that cross-cycle flare forecasting has never been studied.
- Not the first application of SHAP to solar-flare prediction.
- Not a generic Cycle-25 benchmark.
- Not proof that Solar Cycle 25 has different physical laws.
- Not a study in which Cycle 25 is used to retrain the primary model.
- Not a result-seeking project: stability and drift are both acceptable outcomes.

8. Two-Minute Team Version

TEAM SCRIPT

Cycle 24 par model train -> preprocessing, threshold aur model freeze -> independent Cycle 24 holdout aur Cycle 25 temporal blocks par same pipeline test -> TSS/other metrics compare -> same model ke SHAP attributions compare -> within-Cycle-24 sampling + temporal controls aur shift-aware sensitivity checks se verify -> statistically supported conclusion.

Reference Basis

These references support the current design. They are not the final literature review; novelty-sensitive claims must be re-checked before submission.

- [1] Bobra et al. (2014), SHARP data products and HARP/NOAA mapping, Solar Physics. [Source](#)
- [2] Hoeksema et al. (2014), HMI vector magnetic-field pipeline and definitive SHARP products, Solar Physics. [Source](#)
- [3] Chen et al. (2019), SHARP flare precursors and correlated-feature instability, Space Weather. [Source](#)
- [4] Goodwin et al. (2024), solar-cycle and training-window effects in simulated real-time flare forecasting, ApJ. [Source](#)
- [5] Hu et al. (2025), data-source quality issues and Cycle-24-to-25 evaluation, arXiv:2512.13417. [Source](#)
- [6] Wu et al. (2026), interpretable major-flare forecasting with SHAP, MNRAS. [Source](#)
- [7] Li et al. (2026), explainable operational flare forecasting with SHAP, Space Weather. [Source](#)